Tabu search

Lecture 23: Presented by Jacomine Grobler

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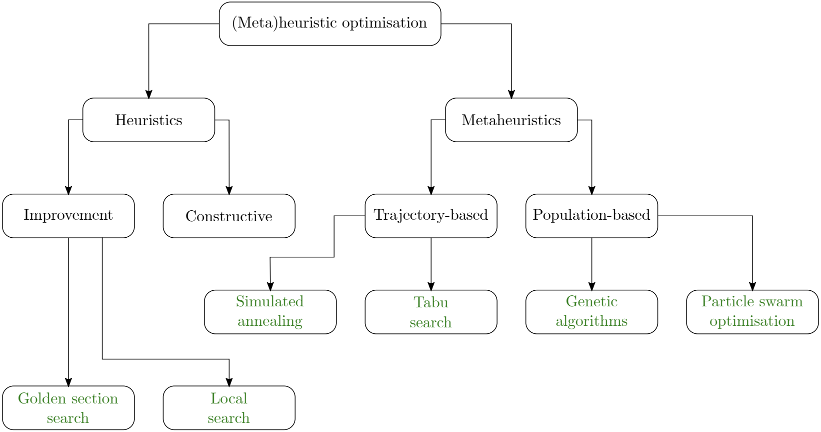
(Meta)heuristics



Heuristics disadvantages

Both golden search and local search heuristics suffer from the same potentially serious disadvantage — they become trapped at local optima in close proximity to the initial conditions of the search, irrespectively of whether or not there are other, higher-quality local optima further away from the starting point(s) of the search.

These heuristics are therefore, in general, incapable of uncovering global optima, except perhaps by chance (*i.e.* if the initial search conditions are selected favourably).

Trajectory-based metaheuristics

A trajectory-based metaheuristic is a generalisation of the notion of local search, equipped with a mechanism for escaping from local optima in order to explore regions of the search space which are not in close proximity to the initial search point.

This escape mechanism differs from one trajectory-based metaheuristic to the next.

Tabu search

Tabu search is a *deterministic* extension of the local search heuristic and adopts a short-term memory structure to resolve the most significant disadvantage of local search — avoiding becoming trapped at poor local optima.

Worsening neighbouring solutions of a current solution can be selected as the next current solution when no improving solutions are available.

While this exception provides a mechanism for escaping from local optima, it unfortunately also introduces the possibility of cycling during the search process.

Certain perturbations are avoided by maintaining a so-called *tabu list* which contains the reversals of recently performed solution perturbations. These perturbation reversals are prohibited so as to avoid cycling during the search.

The tabu list is initialised as an empty list, upon which the reversals of perturbations are inserted into the list as these perturbations are performed.

Each tabu list entry is assigned a fixed number of subsequent search iterations during which its implementation is disallowed called the *tabu tenure* of the perturbation.

Once the required number of search iterations over which implementation of a tabu list entry perturbation was tabu has elapsed, the entry is removed from the tabu list in a bid to reduce its length.

The tabu list is referred to as a *short-term memory* structure.

A *long-term memory* structure is also sometimes employed to promote exploitation of promising areas of the search space in the form of an *aspiration criterion*.

This involves recording in memory the objective function value of the best solution ever encountered during the search (called the *incumbent solution*) and employing the aspiration criterion of overriding the tabu status of any solution perturbation in the tabu list if such a perturbation would lead to a solution with a better objective function value than that of the incumbent solution.

Tabu search

Possible termination criteria:

Stopping after a specified number of search iterations: This termination criterion requires the *a priori* specification of a budget $I_{\max}$ of search iterations that can be afforded within a given computing environment. The algorithm is then terminated once $I_{\max}$ search iterations have been performed.

Stopping when the incumbent solution no longer improves: This termination criterion requires the *a priori* specification of a maximal window of length $W_{\max}$ iterations over which no improvement of the incumbent solution is interpreted as search stagnation. The algorithm is terminated as soon as $W_{\max}$ successive iterations have elapsed since the last improvement of the incumbent solution.

Tabu search

Algorithm 4.4 (Tabu search metaheuristic)

Input: A multivariate function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\boldsymbol{x}^0$ of a local optimum of f, a lattice dimension parameter ε and a termination criterion

Output: An approximation of a local optimum of the function f satisfying the termination criterion

Step 1 $k \leftarrow 0$; Tabu list $\leftarrow \emptyset$; Incumbent $\leftarrow \boldsymbol{x}^0$

Step 2 Generate the lattice neighbourhood of $\boldsymbol{x}^k$ as $\mathcal{N}_\varepsilon(\boldsymbol{x}^k) = \{\boldsymbol{x} \pm \varepsilon \boldsymbol{e}_j,\ j = 1, \ldots, n\}$

Step 3 If $\mathcal{N}_\varepsilon(\boldsymbol{x}^k) = \emptyset$ then output Incumbent and stop

Step 4 $\overline{\boldsymbol{x}}^k \leftarrow$ element of $\mathcal{N}(\boldsymbol{x}^k)$ achieving the best value of f

Step 5 If the perturbation $\boldsymbol{x}^k \rightsquigarrow \overline{\boldsymbol{x}}^k$ is not a member of the tabu list or if the aspiration criterion is satisfied by the perturbation $\boldsymbol{x}^k \rightsquigarrow \overline{\boldsymbol{x}}^k$, then do

Step 6 Insert the reversal of the perturbation $\boldsymbol{x}^k \rightsquigarrow \overline{\boldsymbol{x}}^k$ into the Tabu list and remove any entries from the Tabu list if their tabu tenures have expired

Step 7 $\boldsymbol{x}^{k+1} \leftarrow \overline{\boldsymbol{x}}^k$, $k \leftarrow k + 1$

Step 8 If $f(\overline{\boldsymbol{x}}^k)$ is better than f(Incumbent) then Incumbent $\leftarrow \overline{\boldsymbol{x}}^k$

Step 9 Else $\mathcal{N}(\boldsymbol{x}^k) \leftarrow \mathcal{N}(\boldsymbol{x}^k) \setminus \{\overline{\boldsymbol{x}}^k\}$ and return to Step 3

Step 10 If the termination criterion is satisfied, output Incumbent and stop

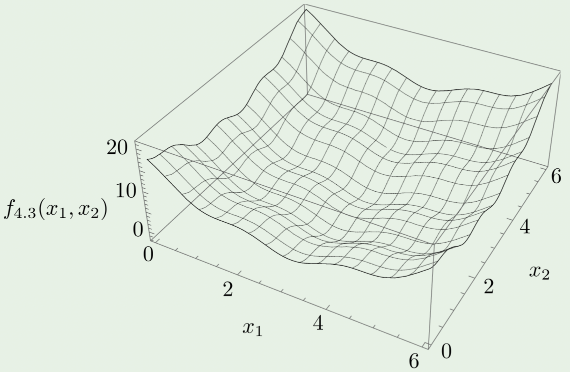
Step 11 Else return to Step 2

Example 4.5 (pp. 125–128)

Minimise the function

$$f_{4.3}(x_1, x_2) = (x_1-3.14)^2+(x_2-2.72)^2+\sin(3x_1+1.41)+\sin(4x_2-1.73)$$

approximately on the domain $(x_1, x_2) \in [0, 6] \times [0, 6]$ by applying the method of tabu search, starting out from the initial estimate $\boldsymbol{x}^0 = [1 \;\; 1]^T$.



k	ε	$\boldsymbol{x}^k$	$\mathcal{N}_\varepsilon(\boldsymbol{x}^k)$	$f_{4.2}(\boldsymbol{x}^k)$	Values of $f_{4.2}$ in $\mathcal{N}_\varepsilon(\boldsymbol{x}^k)$
0	0.3	$\begin{bmatrix}1.0\\1.0\end{bmatrix}$	$\begin{bmatrix}1.3\\1.0\end{bmatrix}, \begin{bmatrix}1.0\\1.3\end{bmatrix}, \begin{bmatrix}0.7\\1.0\end{bmatrix}, \begin{bmatrix}1.0\\0.7\end{bmatrix}$	7.349	6.283, 5.319, 9.317, 8.583
1	0.3	$\begin{bmatrix}1.0\\1.3\end{bmatrix}$	$\begin{bmatrix}1.3\\1.3\end{bmatrix}, \begin{bmatrix}1.0\\1.6\end{bmatrix}, \begin{bmatrix}0.7\\1.3\end{bmatrix}, \begin{bmatrix}1.0\\1.0\end{bmatrix}$	5.319	4.253, 3.880, 7.287, Tabu
2	0.3	$\begin{bmatrix}1.0\\1.6\end{bmatrix}$	$\begin{bmatrix}1.3\\1.6\end{bmatrix}, \begin{bmatrix}1.0\\1.9\end{bmatrix}, \begin{bmatrix}0.7\\1.6\end{bmatrix}, \begin{bmatrix}1.0\\1.3\end{bmatrix}$	3.880	2.814, 3.896, 5.849, Tabu
3	0.3	$\begin{bmatrix}1.3\\1.6\end{bmatrix}$	$\begin{bmatrix}1.6\\1.6\end{bmatrix}, \begin{bmatrix}1.3\\1.9\end{bmatrix}, \begin{bmatrix}1.0\\1.6\end{bmatrix}, \begin{bmatrix}1.3\\1.3\end{bmatrix}$	2.814	2.554, 2.830, Tabu, Tabu
4	0.3	$\begin{bmatrix}1.6\\1.6\end{bmatrix}$	$\begin{bmatrix}1.9\\1.6\end{bmatrix}, \begin{bmatrix}1.6\\1.9\end{bmatrix}, \begin{bmatrix}1.3\\1.6\end{bmatrix}, \begin{bmatrix}1.6\\1.3\end{bmatrix}$	2.554	2.529, 2.569, Tabu, 3.992
5	0.3	$\begin{bmatrix}1.9\\1.6\end{bmatrix}$	$\begin{bmatrix}2.2\\1.6\end{bmatrix}, \begin{bmatrix}1.9\\1.9\end{bmatrix}, \begin{bmatrix}1.6\\1.6\end{bmatrix}, \begin{bmatrix}1.9\\1.3\end{bmatrix}$	2.529	2.127, 2.544, Tabu, 3.967
6	0.3	$\begin{bmatrix}2.2\\1.6\end{bmatrix}$	$\begin{bmatrix}2.5\\1.6\end{bmatrix}, \begin{bmatrix}2.2\\1.9\end{bmatrix}, \begin{bmatrix}1.9\\1.6\end{bmatrix}, \begin{bmatrix}2.2\\1.3\end{bmatrix}$	2.127	1.157, 2.142, Tabu, 3.565
7	0.3	$\begin{bmatrix}2.5\\1.6\end{bmatrix}$	$\begin{bmatrix}2.8\\1.6\end{bmatrix}, \begin{bmatrix}2.5\\1.9\end{bmatrix}, \begin{bmatrix}2.2\\1.6\end{bmatrix}, \begin{bmatrix}2.5\\1.3\end{bmatrix}$	1.157	−0.005, 1.173, Tabu, 2.596
8	0.3	$\begin{bmatrix}2.8\\1.6\end{bmatrix}$	$\begin{bmatrix}3.1\\1.6\end{bmatrix}, \begin{bmatrix}2.8\\1.9\end{bmatrix}, \begin{bmatrix}2.5\\1.6\end{bmatrix}, \begin{bmatrix}2.8\\1.3\end{bmatrix}$	−0.005	−0.703, 0.011, Tabu, 1.434

Example 4.5 (pp. 125–128)

k	ε	$\boldsymbol{x}^k$	$\mathcal{N}_\varepsilon(\boldsymbol{x}^k)$	$f_{4.2}(\boldsymbol{x}^k)$	Values of $f_{4.2}$ in $\mathcal{N}_\varepsilon(\boldsymbol{x}^k)$
14	0.3	$\begin{bmatrix}3.4\\2.8\end{bmatrix}$	$\begin{bmatrix}3.7\\2.8\end{bmatrix}, \begin{bmatrix}3.4\\3.1\end{bmatrix}, \begin{bmatrix}3.1\\2.8\end{bmatrix}, \begin{bmatrix}3.4\\2.5\end{bmatrix}$	−0.788	0.218, −1.553, −0.997, Tabu
15	0.3	$\begin{bmatrix}3.4\\3.1\end{bmatrix}$	$\begin{bmatrix}3.7\\3.1\end{bmatrix}, \begin{bmatrix}3.4\\3.4\end{bmatrix}, \begin{bmatrix}3.1\\3.1\end{bmatrix}, \begin{bmatrix}3.4\\2.8\end{bmatrix}$	−1.553	−0.546, −0.929, −1.761, Tabu
16	0.3	$\begin{bmatrix}3.1\\3.1\end{bmatrix}$	$\begin{bmatrix}3.4\\3.1\end{bmatrix}, \begin{bmatrix}3.1\\3.4\end{bmatrix}, \begin{bmatrix}2.8\\3.1\end{bmatrix}, \begin{bmatrix}3.1\\2.8\end{bmatrix}$	−1.761	Tabu, −1.137, −1.063, Tabu
17	0.3	$\begin{bmatrix}3.1\\3.4\end{bmatrix}$	$\begin{bmatrix}3.4\\3.4\end{bmatrix}, \begin{bmatrix}3.1\\3.7\end{bmatrix}, \begin{bmatrix}2.8\\3.4\end{bmatrix}, \begin{bmatrix}3.1\\3.1\end{bmatrix}$	−1.137	Tabu, 0.485, −0.439, Tabu
18	0.3	$\begin{bmatrix}2.8\\3.4\end{bmatrix}$	$\begin{bmatrix}3.1\\3.4\end{bmatrix}, \begin{bmatrix}2.8\\3.7\end{bmatrix}, \begin{bmatrix}2.5\\3.4\end{bmatrix}, \begin{bmatrix}2.8\\3.1\end{bmatrix}$	−0.439	Tabu, 1.183, 0.723, Tabu
19	0.3	$\begin{bmatrix}2.5\\3.4\end{bmatrix}$	$\begin{bmatrix}2.8\\3.4\end{bmatrix}, \begin{bmatrix}2.5\\3.7\end{bmatrix}, \begin{bmatrix}2.2\\3.4\end{bmatrix}, \begin{bmatrix}2.5\\3.1\end{bmatrix}$	0.723	Tabu, 2.345, 1.692, 0.099
20	0.3	$\begin{bmatrix}2.5\\3.1\end{bmatrix}$	$\begin{bmatrix}2.8\\3.1\end{bmatrix}, \begin{bmatrix}2.5\\3.4\end{bmatrix}, \begin{bmatrix}2.2\\3.1\end{bmatrix}, \begin{bmatrix}2.5\\2.8\end{bmatrix}$	0.099	Tabu, Tabu, 1.068, 0.863
21	0.3	$\begin{bmatrix}2.5\\2.8\end{bmatrix}$	$\begin{bmatrix}2.8\\2.8\end{bmatrix}, \begin{bmatrix}2.5\\3.1\end{bmatrix}, \begin{bmatrix}2.2\\2.8\end{bmatrix}, \begin{bmatrix}2.5\\2.5\end{bmatrix}$	0.863	−0.299, Tabu, 1.833, 1.865

Example 4.5 (pp. 125–128)